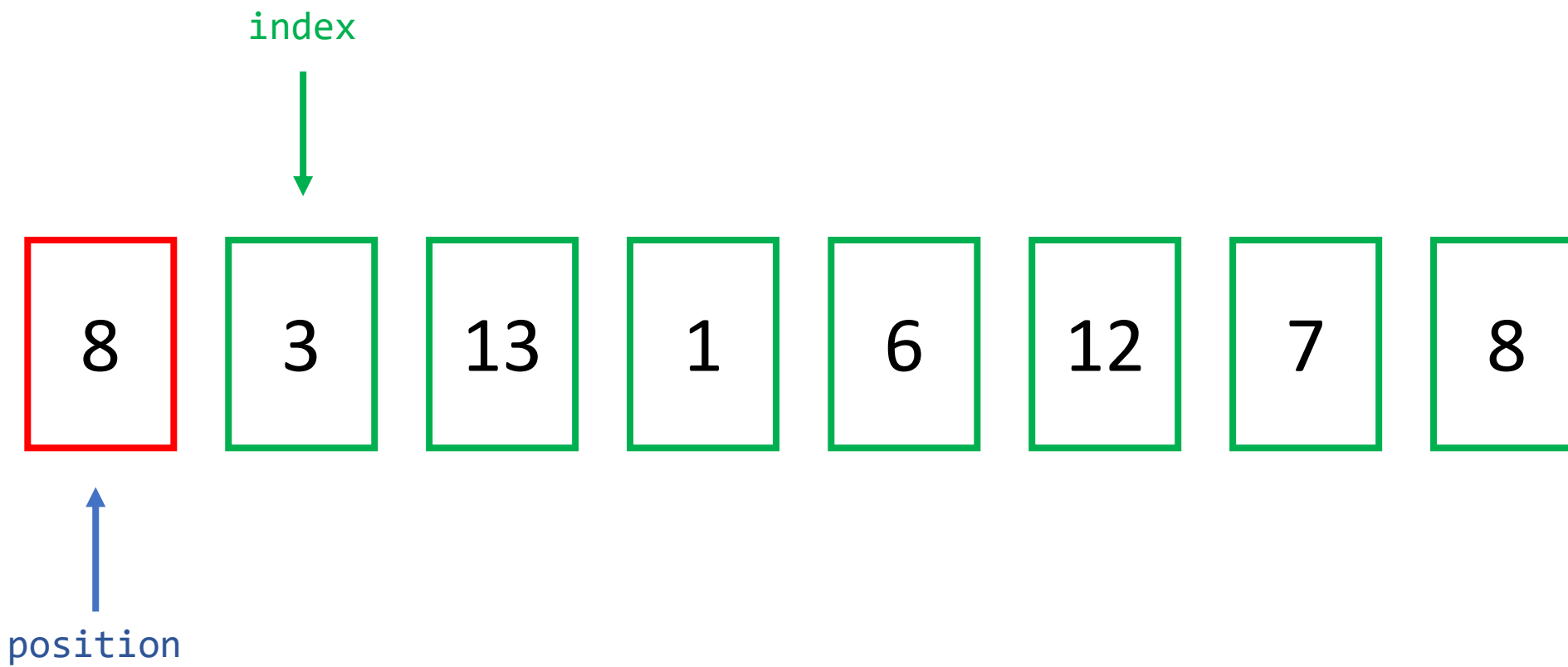


Insertion Sort Animation

```
def insertion_sort(items):  
  
    # Initialise the variables  
    num_items = len(items)  
  
    # Repeat for each item in the list, starting at the second item  
    for index in range(1, num_items):  
  
        # Get the value of the next item to insert  
        item_to_insert = items[index]  
  
        # Get the current position of the last sorted item  
        position = index - 1  
  
        # Repeat while there are still items in the list to check  
        # and the current sorted item is greater than the item to insert  
        while position >= 0 and items[position] > item_to_insert:  
  
            # Copy the value of the sorted item up one place  
            items[position + 1] = items[position]  
  
            # Get the position of the next sorted item  
            position = position - 1  
  
        # Copy the value of the item to insert into the correct position  
        items[position + 1] = item_to_insert
```



index = 1

position = 0

Insert item into correct place in sorted list

Is $8 > 3$?

Yes, copy 8 to position + 1

3

8

13

1

6

12

7

8

↑
position

index = 1

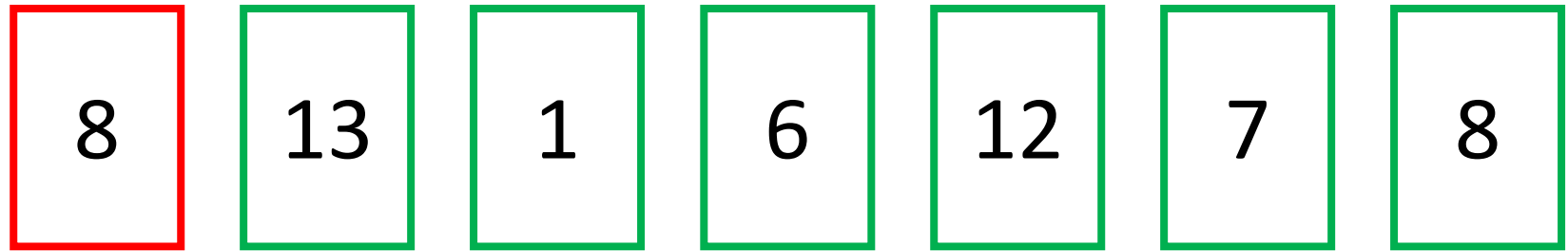
position = 0

Insert item into correct place in sorted list

Is $8 > 3$?

Yes, copy 8 to position + 1

3



↑
position

index = 1

position = 0

Insert item into correct place in sorted list

`position < 0`

Insert 3 at position 0

3

8 13 1 6 12 7 8

↑
position

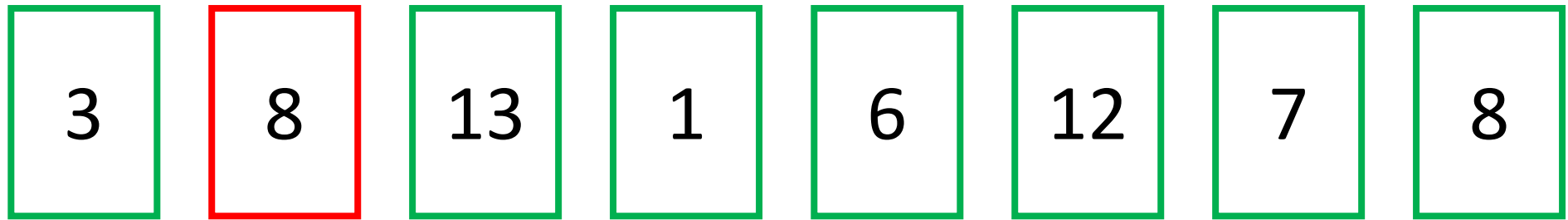
`index = 1`

`position = -1`

Insert item into correct place in sorted list

position < 0

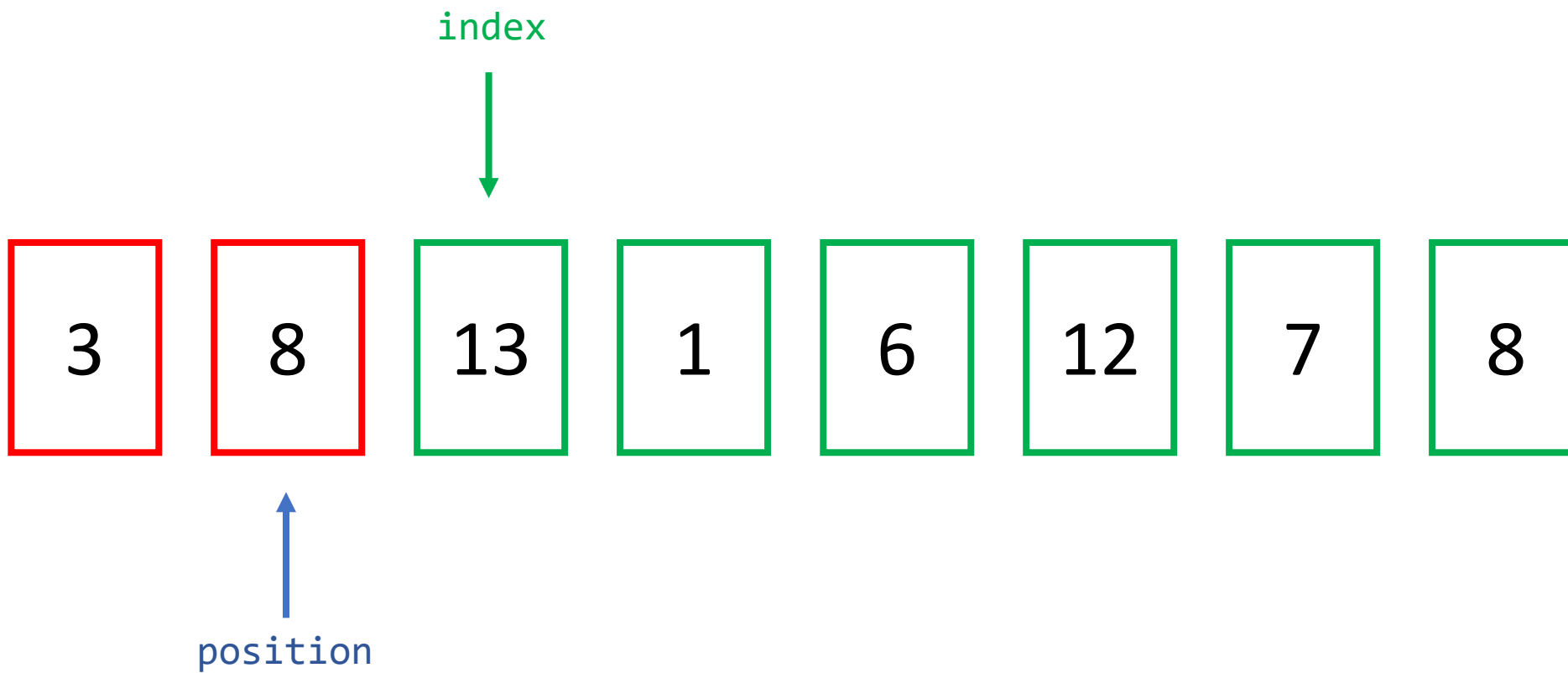
Insert 3 at position 0



↑
position

index = 1

position = -1



index = 2

position = 1

Insert item into correct place in sorted list

Is $8 > 13$?

No, insert 13 at position + 1

13

3

8

1

6

12

7

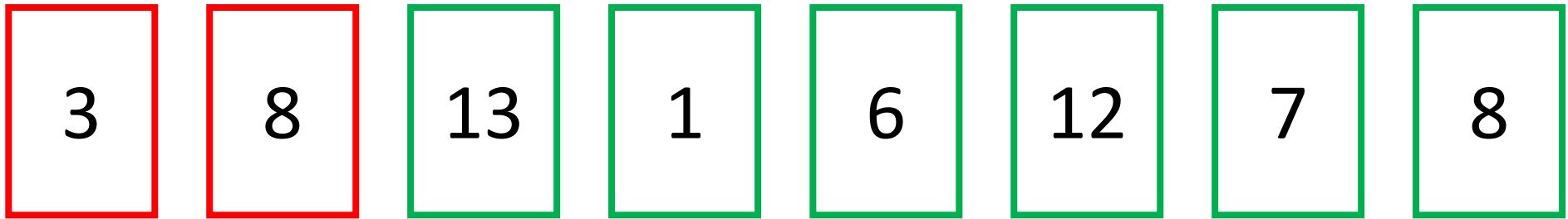
8

↑
position

Insert item into correct place in sorted list

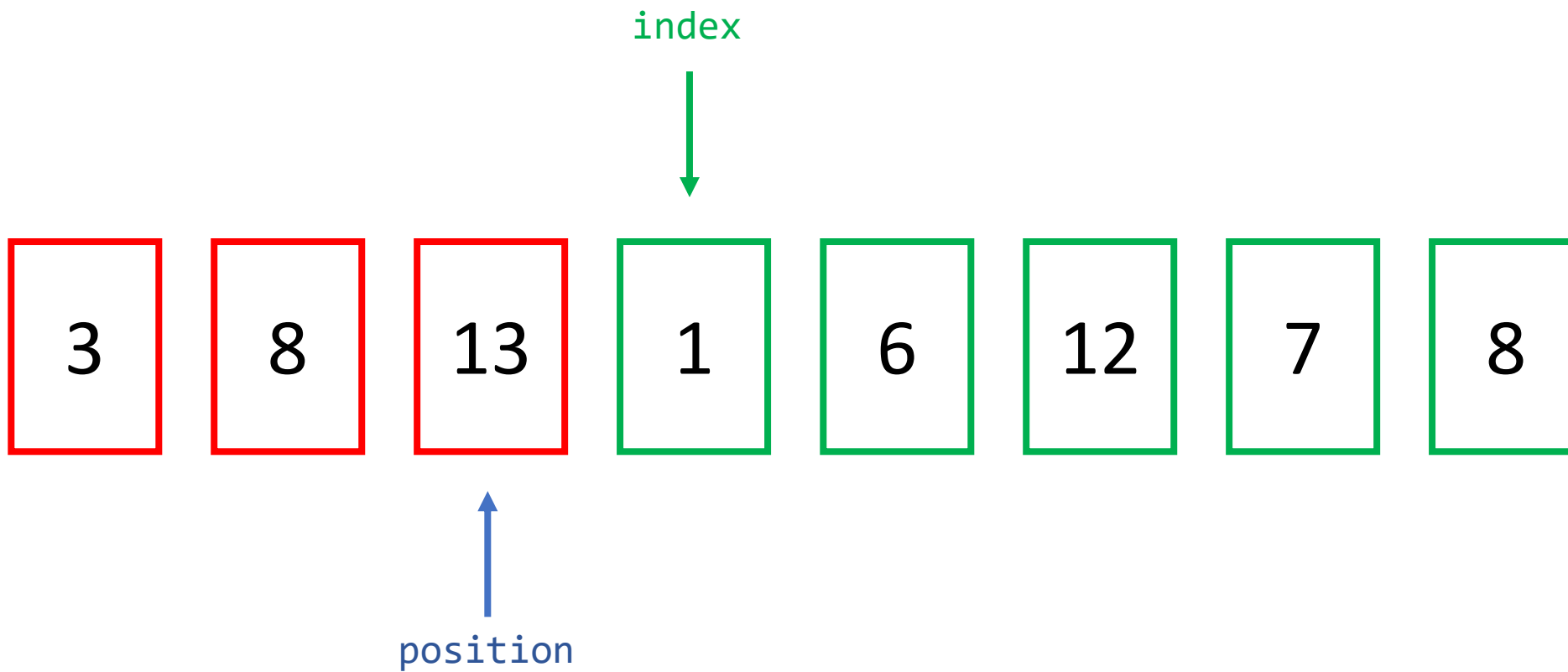
Is $8 > 13$?

No, insert 13 at position + 1



index = 2

position = 1



index = 3

position = 2

Insert item into correct place in sorted list

Is $13 > 1$?

Yes, copy 13 to position + 1

1

3

8

13

6

12

7

8

↑
position

index = 3

position = 2

Insert item into correct place in sorted list

Is $13 > 1$?

Yes, copy 13 to position + 1

1

3

8

13

6

12

7

8

↑
position

index = 3

position = 2

Insert item into correct place in sorted list

Is $8 > 1$?

Yes, copy 8 to position + 1

1

3

8

13

6

12

7

8

↑
position

index = 3

position = 1

Insert item into correct place in sorted list

Is $8 > 1$?

Yes, copy 8 to position + 1

1

3

8

13

6

12

7

8

↑
position

index = 3

position = 1

Insert item into correct place in sorted list

Is $3 > 1$?

Yes, copy 3 to position + 1

1

3

8

13

6

12

7

8

↑
position

index = 3

position = 0

Insert item into correct place in sorted list

Is $3 > 1$?

Yes, copy 3 to position + 1

1

3

8

13

6

12

7

8

↑
position

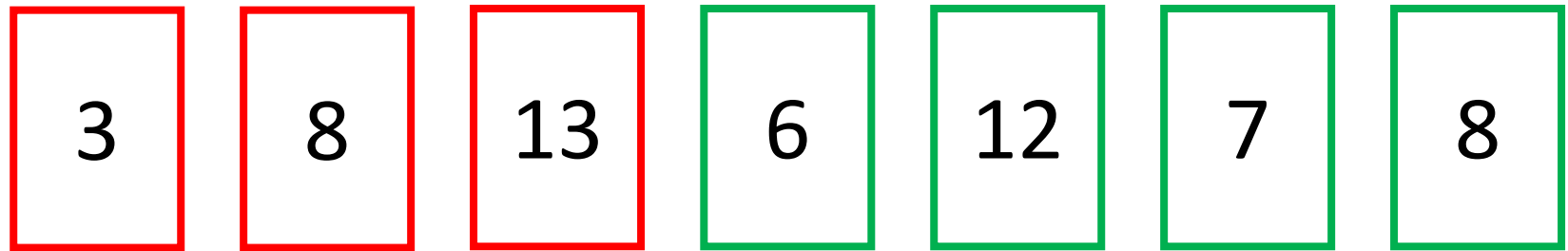
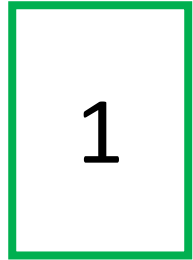
index = 3

position = 0

Insert item into correct place in sorted list

`position < 0`

Insert 3 at position 0



↑
position

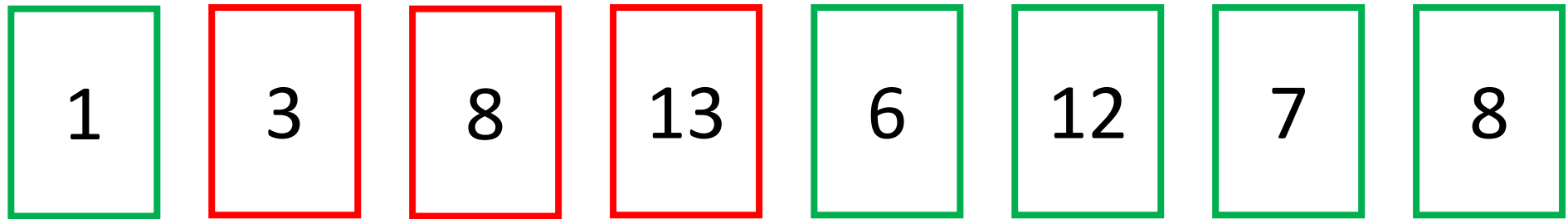
`index = 3`

`position = -1`

Insert item into correct place in sorted list

`position < 0`

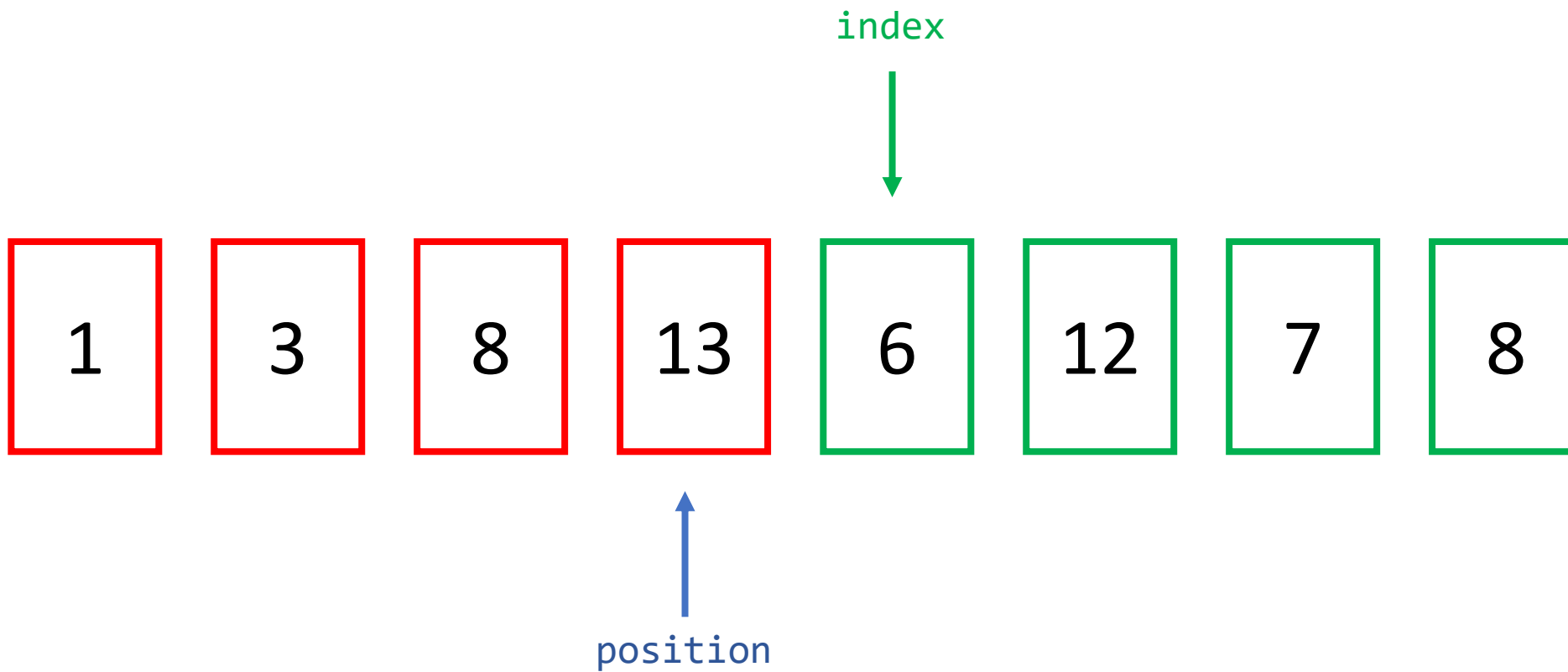
Insert 3 at position 0



↑
position

`index = 3`

`position = -1`



index = 4

position = 3

Insert item into correct place in sorted list

Is $13 > 6$?

Yes, copy 13 to position + 1

6

1 3 8 13

12 7 8

↑
position

index = 4

position = 3

Insert item into correct place in sorted list

Is $13 > 6$?

Yes, copy 13 to position + 1

6

1

3

8

13

12

7

8

↑
position

index = 4

position = 3

Insert item into correct place in sorted list

Is $8 > 6$?

Yes, copy 13 to position + 1

6

1

3

8

13

12

7

8

↑
position

index = 4

position = 2

Insert item into correct place in sorted list

Is $8 > 6$?

Yes, copy 13 to position + 1

6

1

3

8

13

12

7

8

↑
position

index = 4

position = 2

Insert item into correct place in sorted list

Is $3 > 6$?

No, insert 6 at position + 1

6

1

3

8

13

12

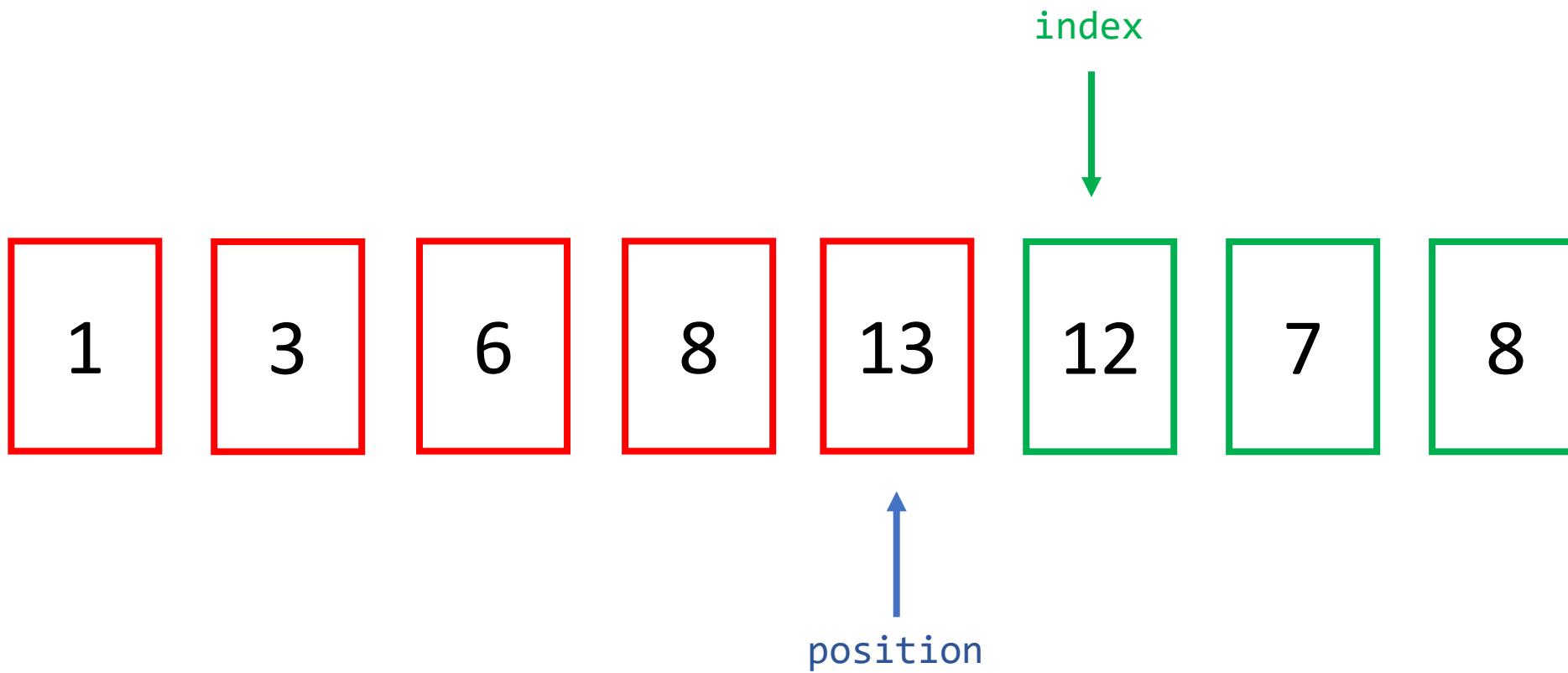
7

8

↑
position

index = 4

position = 1



index = 5

position = 4

Insert item into correct place in sorted list

Is $13 > 12$?

Yes, copy 13 to position + 1

12

1

3

6

8

13

7

8

↑
position

index = 5

position = 4

Insert item into correct place in sorted list

Is $13 > 12$?

Yes, copy 13 to position + 1

12

1

3

6

8

13

7

8

↑
position

index = 5

position = 4

Insert item into correct place in sorted list

Is $8 > 12$?

No, insert 12 at position + 1

12

1

3

6

8

13

7

8

↑
position

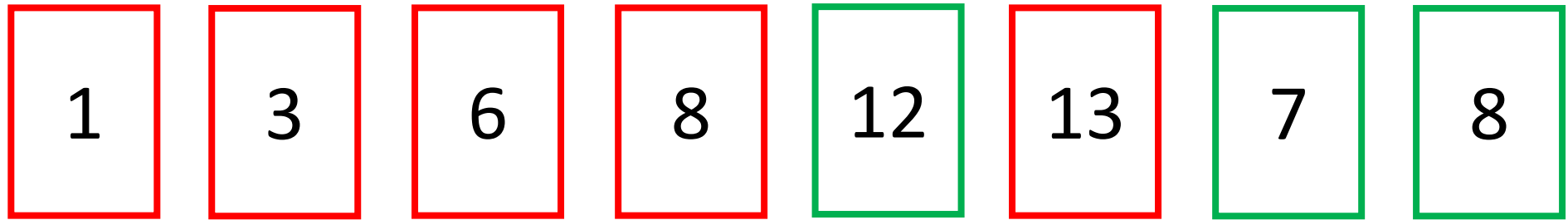
index = 5

position = 3

Insert item into correct place in sorted list

Is $8 > 12$?

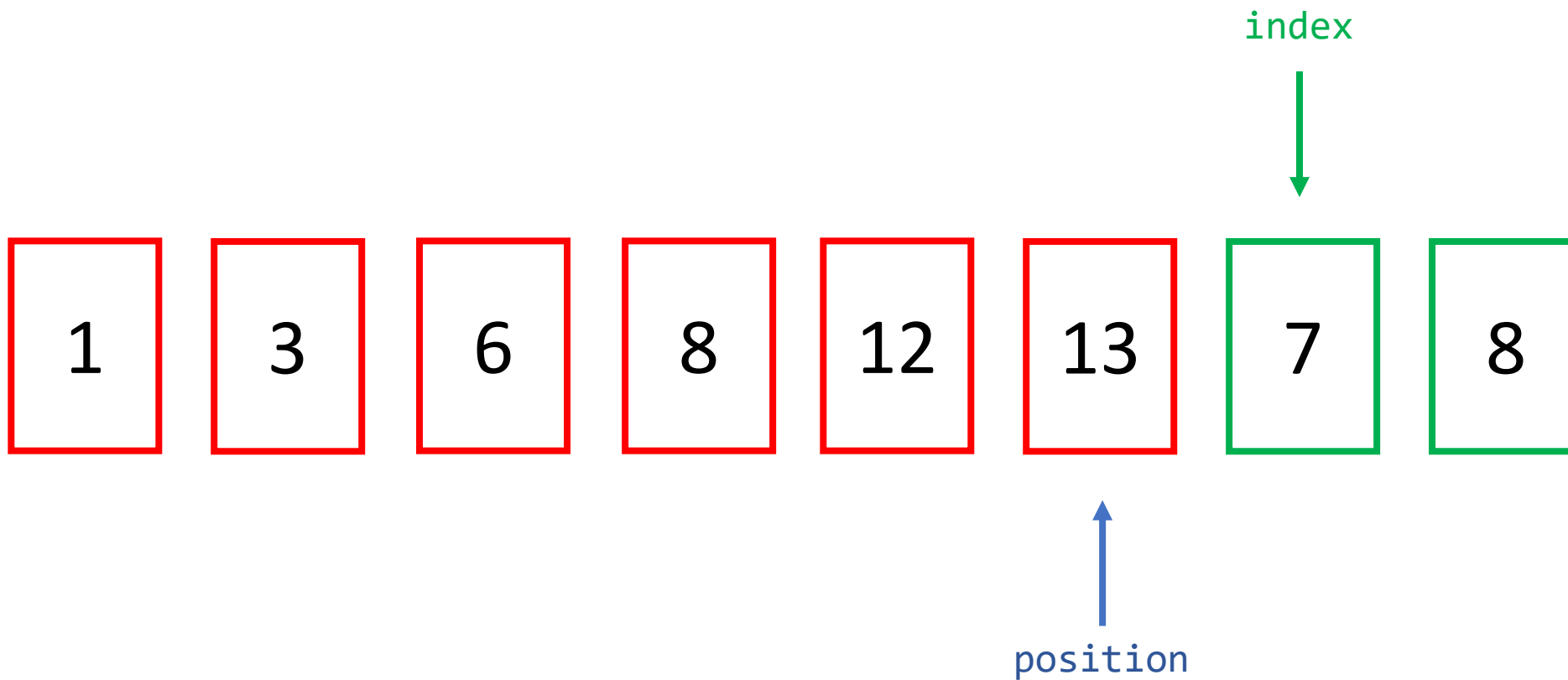
No, insert 12 at position + 1



position

index = 5

position = 3



index = 6

position = 5

Insert item into correct place in sorted list

Is $13 > 7$?

Yes, copy 13 to position + 1

7

1

3

6

8

12

13

8

↑
position

index = 6

position = 5

Insert item into correct place in sorted list

Is $13 > 7$?

Yes, copy 13 to position + 1

7

1

3

6

8

12

13

8

↑
position

index = 6

position = 5

Insert item into correct place in sorted list

Is $12 > 7$?

Yes, copy 12 to position + 1

7

1

3

6

8

12

13

8

↑
position

index = 6

position = 4

Insert item into correct place in sorted list

Is $12 > 7$?

Yes, copy 12 to position + 1

7

1

3

6

8

12

13

8

↑
position

index = 6

position = 4

Insert item into correct place in sorted list

Is $8 > 7$?

Yes, copy 8 to position + 1

7

1

3

6

8

12

13

8

↑
position

index = 6

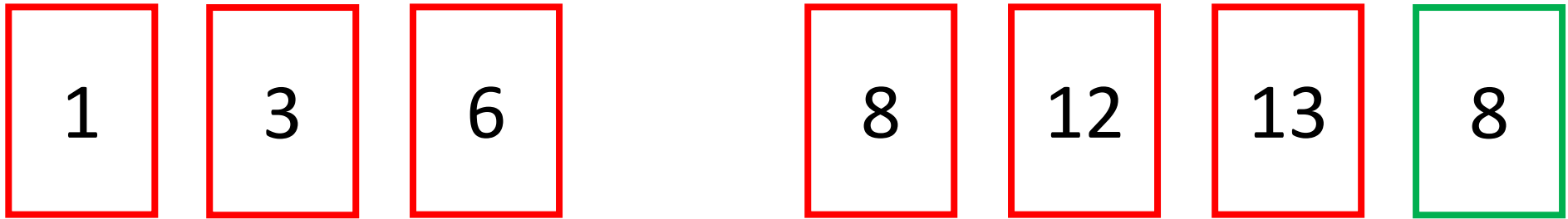
position = 3

Insert item into correct place in sorted list

Is $8 > 7$?

Yes, copy 8 to position + 1

7



position

index = 6

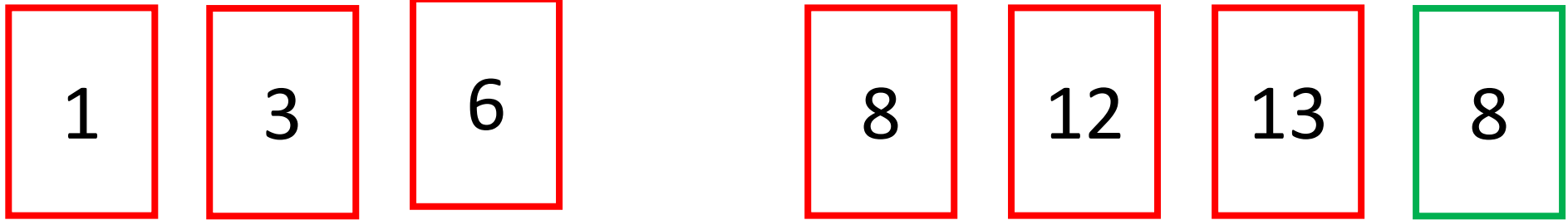
position = 3

Insert item into correct place in sorted list

Is $7 > 6$?

Yes, insert 7 at position + 1

7



position

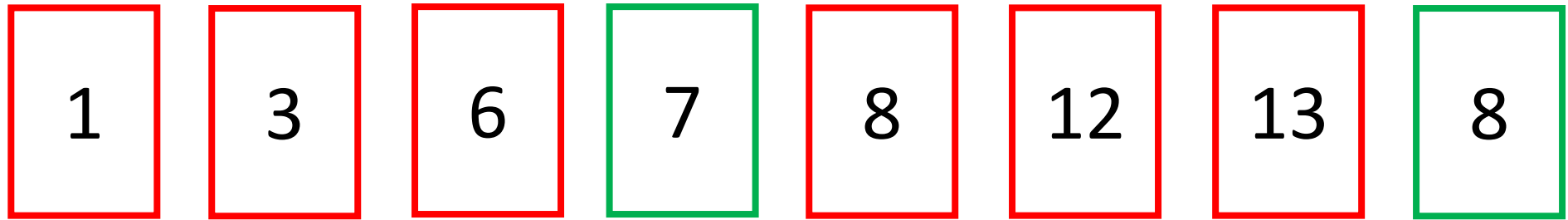
index = 6

position = 2

Insert item into correct place in sorted list

Is $7 > 6$?

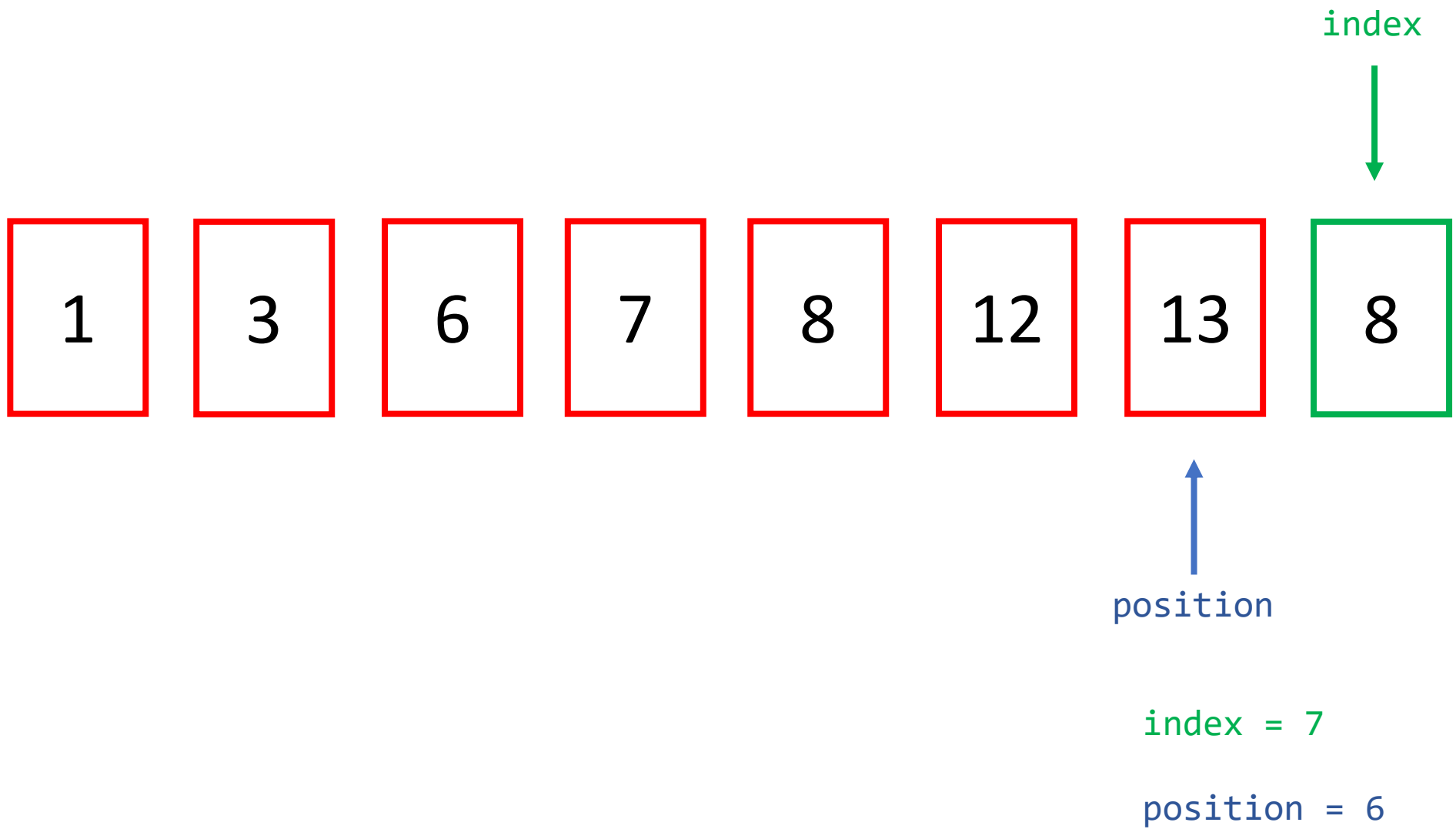
Yes, insert 7 at position + 1



position

index = 6

position = 2

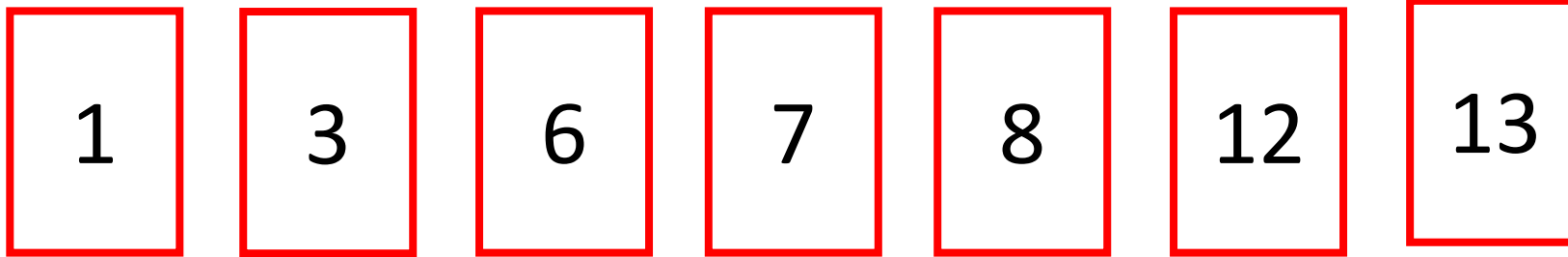


Insert item into correct place in sorted list

Is $13 > 8$?

Yes, copy 13 to position + 1

8



↑
position

index = 7

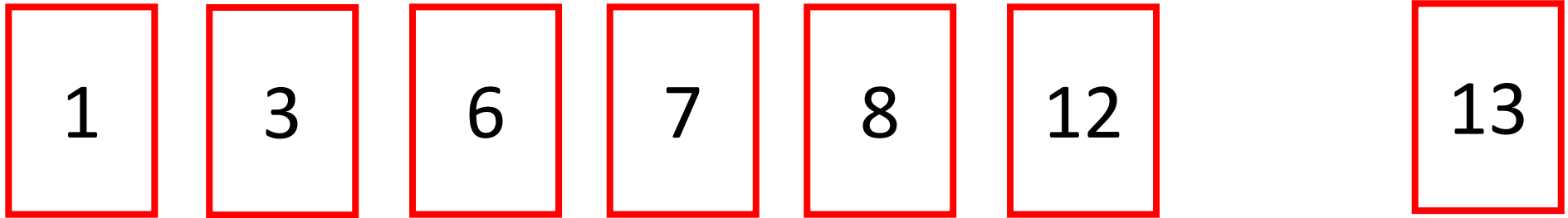
position = 6

Insert item into correct place in sorted list

Is $13 > 8$?

Yes, copy 13 to position + 1

8



position

index = 7

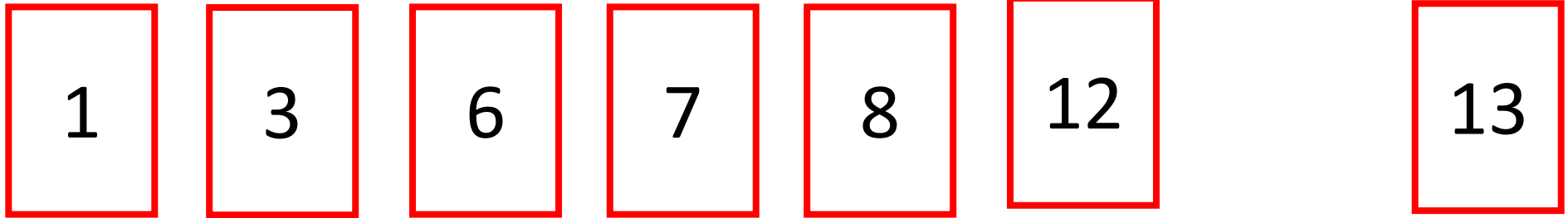
position = 6

Insert item into correct place in sorted list

Is $12 > 8$?

Yes, copy 12 to position + 1

8



index = 7

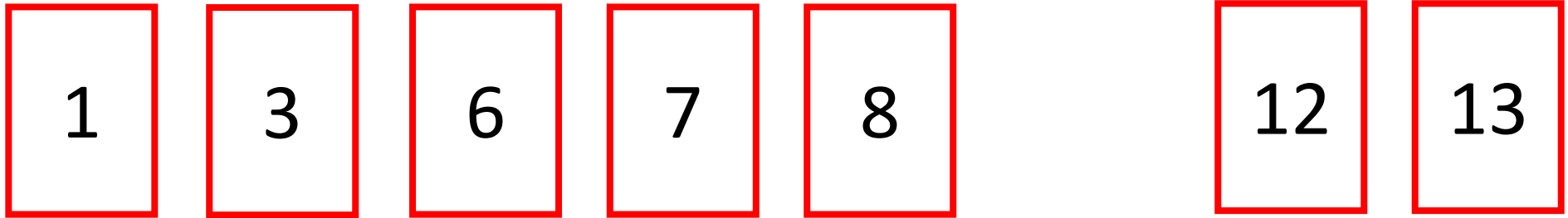
position = 5

Insert item into correct place in sorted list

Is $12 > 8$?

Yes, copy 12 to position + 1

8



position

index = 7

position = 5

Insert item into correct place in sorted list

Is $8 > 8$?

No, insert 8 at position + 1

8

1

3

6

7

8

12

13

↑
position

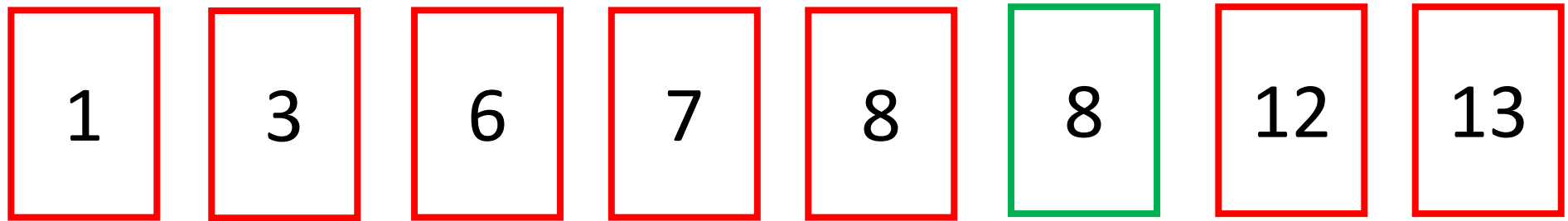
index = 7

position = 4

Insert item into correct place in sorted list

Is $8 > 8$?

No, insert 8 at position + 1



↑
position

index = 7

position = 4

